

Design and Implementation of a Smart Tele-Controlled IV System for Reverse Blood Flow Prevention and Infusion Pump-Based Drug Delivery

BME 4232: Telemedicine and Health Care Laboratory



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Abstract

Intravenous (IV) therapy plays an important role in healthcare, which allows doctors to put medications straight into a patient's bloodstream. One of the major challenges in IV therapy is the occurrence of reverse blood flow. It can contaminate the blood, make treatments fail, and cause health issues like blood clotting and loss of blood. Manual infusion methods for drug delivery may lead to dosage errors and inconsistency. So, accurate and timely drug delivery is very important. This project aims to create an IoT-based smart wearable IV cannula system to spot and stop backward flow in real time and a controlled and automated drug delivery system through an integrated infusion pump. The system integrates an LED-LDR sensor module pair for detecting saline and a mini solenoid valve to automatically stop the reverse blood flow when detected. The system uses an ESP32-S2 Mini and a ESP32 microcontroller as the central processing unit, which are also integrated with wi-fi module enabling seamless communication with a digital interface. Real-time updates of the patient are displayed on an OLED with additional notifications shown through an app. The project also incorporates an infusion pump that delivers the drugs in a controlled and automated way. The system has the MAX30102 sensor to check the patient's real-time vital signs, like heart rate and oxygen saturation (SpO2). The ESP32 and servo motor control the infusion pump and give precise doses of drugs in a controlled way. So, doctors or nurses can easily enter the right amount of drugs and control delivery through a web interface as well as enter it with manual keypad on the device. This low-cost and effective solution tries to improve the safety of IV therapy for backflow prevention and automatic drug delivery by reducing caregiver tasks and continuous attention. The system can be a scalable solution for resource-limited hospitals and support remote monitoring of infectious diseases such as COVID-19.

List of Abbreviations & Symbols

DC-DC	Intravenous Direct Current to Direct Current
HR	Heart Rate
IV	Intravenous
LED	Light Emitting Diode
LDR	Light Dependent Resistor
Li-Po	Lithium Polymer
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
SpO ₂	Peripheral Capillary Oxygen Saturation

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Chapter 1

Introduction

1.1 Background

The word Intravenous refers to “within the vein”. Intravenous (IV) therapy is a type of treatment in which fluids are delivered directly into the bloodstream through veins [1]. This is a common, affordable procedure in the treatment of different diseases. In most cases, the injected fluid is saline, crystalloids, and blood products. The products are mainly deployed to provide external energy, prevent dehydration, heal during or after any operation, and to maintain the amount of water, salt, and other necessary body elements [2].

An IV setup consists of a drip bottle or IV bag that contains the fluid, a drip chamber to monitor fluid flow, a roller clamp to control the flow rate, and a capillary tube called an IV tube. The IV bag, which contains the fluid, is placed at a height above the patient. As a result, a pressure difference can be observed between the fluid in the bag and the patient’s venous pressure. This pressure difference allows the fluid to flow into the bloodstream. [3] Whenever the level of fluid drops becomes low, a reverse pressure can be noticed. This reverse pressure allows human blood cells to flow backward and move towards the IV bag. [4] The backflow of blood may result in serious complications, including thromboembolism in the heart or lungs, which may lead to death. To prevent this backflow, multiple approaches have been introduced over the years.

Advanced medical systems have implemented multiple techniques to prevent the backflow of blood in the IV cannula system. The solutions have been developed using alarm-based systems, RFID-based systems, Optical detection, Flow rate sensors, One-way valves/non-return valves, and Wireless sensors [5].

The proposed system presents a smart IV cannula design and implementation using a solenoid valve to effectively prevent blood backflow. The system is integrated with the ESP32-S2 Mini web server, enabling real-time monitoring and transmission of physical parameters along with the solenoid valve status through a wireless interface. Users can remotely monitor and control the valve operation through the web-based platform.

Automation in medical procedures has seen considerable uptake in recent decades, driven by advances in IoT and sensor technologies. An important aspect of medical treatment is drug delivery, classically carried out by skilled healthcare professionals who calculate and dispense the exact dosage of the drug. Manual administration may be inconsistent due to human error, time constraints, and fatigue; hence, the use of IoT-based solutions poses an alternative in providing real-time monitoring and control to enhance safety and precision.

In a traditional drug delivery system, the healthcare provider must keep a constant eye on the patient and modulate the dosage accordingly. This dependence on manual intervention is both time-consuming and prone to errors. A recent improvement in IoT has enabled the creation of systems that can act upon real-time data from sensors. Such systems will increase the reliability of devices used in medicine and reduce the workload of health professionals.

Furthermore, the affordable availability of low-cost microcontrollers, sensors, and wireless communication modules is making it possible to implement IoT-based solutions in resource-constrained healthcare settings. Combining these technologies, the project aims at developing a system that will ensure adequate and precisely controlled infusion delivery system.

1.2 Literature Review

The blood backflow problem in IV therapy was noticed a long time after the first clinical application of IV therapy. Then the researchers started to find different methods to overcome this serious complication. The improvement of the methods was noticeable later on. The early Intravenous therapy was done in 1976 on 130 patients successfully, which continued for two days [6]. No patient faced any type of serious blood infection (septicaemia) during the procedure, although 48% IV cannulas had bacteria. Dychter et al [7] have identified the blood backflow as a significant contributing factor to catheter failure. The study provided the economic justification for developing the backflow prevention systems. Bhavasaar et al. [1] have proposed an automated intravenous fluid monitoring and alerting system in which an alarm and LED indicator were placed in the nurse station for alert. A load cell was used to monitor the fluid level, and a heartbeat sensor was used to detect air embolisms. Shelishiyah et al. [8] recommended a voice alert system that was designed with an IR sensor and a microcontroller. A microwave reflectometry-based system was presented by Cataldo et al. [9] that can control and monitor the intravenous drip infusions, and thus prevention of the backflow of blood can be achieved. An apparatus was developed to give an automatic warning signal once the IV fluid runs out using the RFID technology, according to Huang et al. [10]. An automated locking system was suggested by Moorthy et al. [4] for preventing backflow of blood with the use of a microprocessor, a load cell, and a solenoid valve in order to weigh the IV bags and lock the tube automatically when the fluid level goes below a certain limit. In the year 2024, Jants et al. introduced a smart IV catheter system capable of monitoring various physiological factors.

Drug delivery systems are important for effective medical treatment because they ensure that drugs are delivered with the correct amount and at the exact locations within the human body. As per S. M. R. Islam et al. (2015), drug delivery can be facilitated effectively using technology solutions, which include IoT devices that can monitor and control the process in real-time [11]. The intravenous method of drug delivery is considered one of the most common because of the quick action and effectiveness associated with this mode. But according to Linda T. Kohn et al. (2000), the problem of medication errors related to dose and rate is prevalent in healthcare systems [12].

K.N Baluprithviraj Et al. [13] proposed an IoT-based system that use a load cell to gauge the level of the saline bottle and a solenoid valve in order to stop the opposite flow of blood and displays the data on a mobile app using Blynk platform.

S Barath Sanjay Et al. [14] proposed a prototype of Smart IV Drip Stand that employs a load cell to determine the level of the IV fluid and a solenoid valve to prevent the blood's opposite flow.

The prototype also uses a keypad and a servomotor to adjust the flow rate manually or automatically. The paper presents both software and hardware design, as well as the experimental results of the prototype.

Rajasekar Rathanasamy Et al. [15] propose a smart device to regulate the blood flow from the intravenous line in reverse, making use of a storage cylinder, an automated flow-limiting valve, and a specially made unidirectional valve to prevent the blood's opposite flow.

In contrast to these existing solutions, our proposed system integrates the IV and drug delivery system, which can be both manually and automatically controlled, reducing the limitations to a great extent.

1.3 Objectives

The primary objective of this project is to create a low-cost, automated, and remotely operated IV system that provides backflow protection and a controlled infusion system. This research intends to:

- Design and implement a method for detecting and preventing reverse flow of blood within the IV system.
- Develop an automated system to accurately control drug infusion.
- Create an IoT-based interface for continuous monitoring and controlling the integrated system from a remote location.

Chapter 2

Methodology

2.1 Overview of the Proposed System

The flowchart of the proposed Smart Tele-Controlled Intravenous Drug Delivery and Reverse Blood Flow Prevention System is shown in Fig. 2.1. It illustrates the integrated operation of the dual-ESP32 architecture, consisting of the Smart IV Monitoring Unit and the Smart Infusion System Unit. The telemedicine interface acts as the central monitoring and control platform, which enables real-time visualization, status updates, and remote operation of both subsystems.

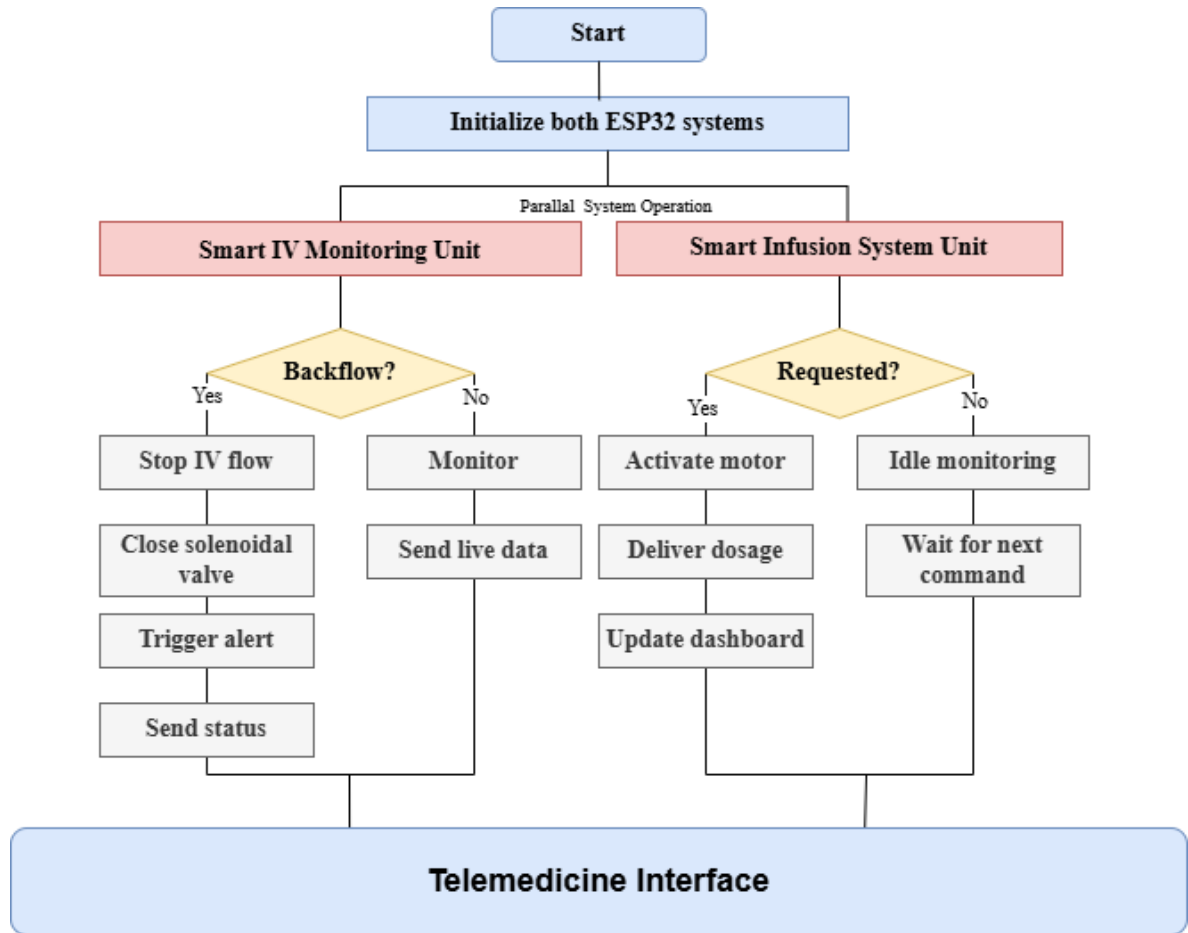


Fig. 2.1: Overall flowchart of the proposed system

2.2 Working Principle of Backflow Detection

Beer–Lambert Law

The optical sensing principle is governed by the Beer–Lambert Law, which defines the exponential attenuation of light as it passes through an absorbing medium:

$$I = I_0 e^{-x\mu} \dots\dots\dots(1)$$

Where,

I_0 : Incident light intensity from the LED source

I : Transmitted light intensity detected by the sensor

μ : Wavelength-dependent absorption coefficient of the medium (saline or blood)

x : Optical path length corresponding to IV tubing diameter

This relationship indicates that variations in fluid composition directly affect transmitted light intensity, enabling material differentiation [15].

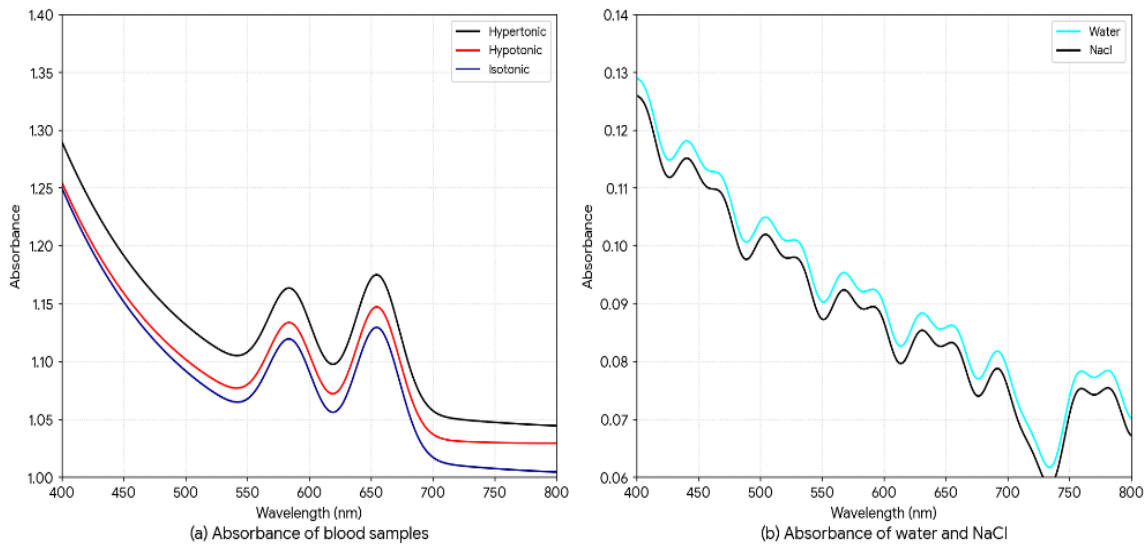


Fig. 2.2: Light Absorption Characteristics of Blood and Saline.

The suggested method involves the use of an optical sensor technology to continuously monitor the intravenous (IV) fluid flow line, with detection based on any changes in the level of optical attenuation, which is due to the differing levels of absorption in either saline or blood.

Optical Characteristics and Wavelength Optimization

The method uses the difference in spectral absorption between saline and blood. Saline demonstrates weak optical absorption at wavelengths from the visible light region, with typical absorbances being less than 0.14. Hence, it provides a maximum transmission of light through the IV catheter, forming a good reference point. Blood shows dependence on wavelength for absorbance properties due to the presence. One of the characteristic absorption peaks can be observed within the blue part of the electromagnetic spectrum (400–450 nm), as shown in Fig. 2.2. This band is referred to as a Soret band. Two secondary Q-bands may also be observed at 540 nm and 575 nm [16]. To maximize detection sensitivity, the system employs a light source within the 400–450 nm range. At this wavelength, the absorption coefficient (μ) of blood is significantly

higher than that of saline, producing a pronounced reduction in transmitted intensity when blood is present [17]. The optical signal is converted into an electrical signal using a Light Dependent Resistor (LDR) configured in a voltage divider circuit. The resistance of the LDR (R_{LDR}) varies inversely with incident light intensity (I).

The resulting voltage output (V_{out}) is given by:

$$V_{out} \propto R_{LDR}(I)$$

The increase in optical absorption during blood backflow is known as the Soret band effect. It leads to higher LDR resistance, which results in an increase in V_{out} . Once V_{out} exceeds the preset value, the alarm and other mechanical actions, such as closing the solenoid valve, are set into motion [18]. Table 2.1 shows the LDR-based variation under different flow conditions.

Table 2.1: LDR-Based Signal Variation Under Different Flow Conditions

Condition	Light Intensity	LDR Resistance	Voltage	System Action
Normal Flow	High	Low	Low	Continue
Blood Backflow	Low	High	High	Trigger alert + close

2.3 Hardware of the Smart IV System

S2 Mini V1.0.0 ESP32

The compact and powerful ESP32-S2-based development board S2 Mini V1.0.0 is designed for IoT applications. It has a well-grounded and stable ESP32-S2FN4R2 Wi-Fi connection. This board is an evolution of D1 mini, which was based on ESP8266. There are many more GPIO and many more functionalities in the board. It has built-in memory of 4MB Flash and 2MB PSRAM, which provides sufficient memory and storage capacity. It has 320KB of inbuilt memory in the form of SRAM and 128KB of ROM memory for boot loading and system functions. Also has a Type-C USB interface for versatile and efficient connectivity. It features 27 GPIO pins with a number of different functions as interrupt, PWM, 12C, single wire ADC, DAC, SPI, UART, and USB OTG. The dimension of this board is 34.3×25.4mm with a weight of 2.4g. The Arduino IDE can be used to upload the code to this board.

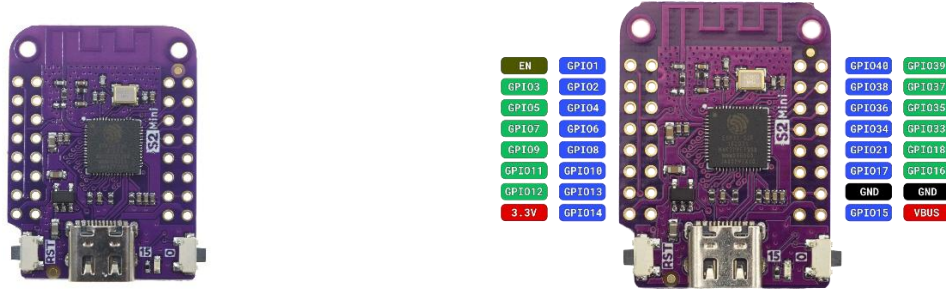


Fig. 2.3: ESP32 Wi-Fi Module

Mini Solenoidal Valve

A mini solenoidal valve is a compact, electromechanical device that is made for controlling the flow of fluids or gases accurately. When an electric current is applied to the coil, it produces a magnetic field that turns on a plunger to open or close the valve orifice based on the electromagnetic principle. Good for low-volume flow in biomedical devices, portable analytical, and microfluidic devices. The benefits of these valves include low power usage, quick response times, and reliability. These valves, combined with an Arduino microcontroller such as Arduino Uno, Arduino Mega, or ESP32, create an accurate and automatic control in smart systems.

N-Channel MOSFET (IRFZ44N)

IRFZ44N is a widely used N-channel enhancement-mode power MOSFET designed for high-efficiency switching applications. It has low state resistance and high current handling capabilities, suitable for applications in power electronics, motor control, and embedded systems. Best suited for medium/high power switching applications. It has a rapid switching speed and a low gate charge, so it's suitable for applications such as pulse width modulation, DC-DC conversion, solenoid drive, and low voltage motor drive. IRFZ44N acts as an effective interface when the current load is high, with a microcontroller.

LDR Sensor Module (LM393)

LDR Sensor Module LM393 is a combination of an LDR sensor module and LM393 voltage comparator on a 33×14×8mm PCB. It operates on 3.3-5V and offers analog and digital outputs with an onboard threshold potentiometer, sub-microsecond response, and 15mA drive facility ideal for automated lighting, ambient light feedback, and energy-efficient control systems. In real-world scenarios, automatic switching can be achieved via interfacing with microcontrollers or by directly controlling relay modules. It gives real-time diagnostics in system development and deployment. This module is easy to install and includes holes and signals to facilitate quick installation in prototype as well as production environments.



Fig. 2.4: (a) Mini Solenoidal Valve



Fig. 2.4: (b) N-Channel MOSFET (IRFZ44N)



Fig.2.4: (c) LDR Sensor Module (LM393)

LED (Blue)

LED, Light Emitting Diode – a semiconductor that releases light when a current flows through it. It is also energy efficient, flexible, durable, and resistant to vibration and impact damage. Can be very small and has flexible design options. Very durable and seldom burns out.

Pulse and Heart Rate Sensor (MAX 30102)

The MAX30102 Heart Rate and Pulse Oximeter Sensor Module is a high-quality biosensor designed for use in wearable sensing devices and monitoring systems. It has incorporated red and infrared LEDs for effective biosignal detection. Authentic Reading occurs when external and internal light sources are reduced. Excellent data communication with a photo detector and a compatible standard 12C interface. It has a large voltage range and has almost zero power consumption when not in use. This module ensures measurement of heart rate and blood oxygen level. It can be incorporated into fitness trackers, wearable health monitors, smartwatches, fitness bands, and other portable gadgets. The module is a reliable and efficient real-time health monitoring solution.

Rechargeable Li-Po Battery (2000 mAh)

A lithium polymer battery is a small and light battery. It has a nominal voltage of 3.7V and a capacity of 2000mAh. Typically employed in portable devices and in embedded systems. It has an integrated protection circuit to ensure short-circuit, over-charge, and over-discharge protection, plus more than 500 charge/discharge cycles with high energy density. The battery comes with a capacity of 60x40x8mm and has a reliable and stable output. It can be charged with standard Li-Po chargers, making it ideal for robotics, IoT, or DIY projects.



Fig. 2.5: (a) Pulse and Heart Rate Sensor (MAX 30102)



Fig. 2.5: (b) Rechargeable Li-Po Battery (2000 mAh)

DC-DC Step-Up Boost Module (MT3608)

MT3608 is an adjustable, effective DC-DC boost converter Module. It's designed to boost the small voltage DC input to a fixed higher voltage output. It has a peak current of 2A and can reach a maximum efficiency of 93%. It is based on a 1.2 MHz controller and an integral trimpot to provide precise output tuning. This module is suitable for portable and embedded applications. It offers a low-rippled output and offers a lot of protection features, including under-voltage lockout, current limiting, and thermal shutdown. It can be used in a variety of battery-operated and low-voltage electronic applications requiring step-up voltage regulation. This module offers the solution for power control of devices with power requirements that need to be raised, including mobile electronics, do-it-yourself power supplies, and distance sensors.

Adjustable Step Down Module (MP1584EN)

The MP1584EN is an adjustable buck (down) DC-DC conversion module. It can take in the power of 4.5-28V and output up to 20V at up to 3A with an efficiency of 95% or higher. It features high frequency, current-mode switching with user-adjustable voltage regulation and built-in protection. Ideal for applications where there is limited space and high efficiency requirements with electronics.

Thermistor

A thermistor is a kind of semiconductor resistor. Its resistance highly depends on temperature. They have a high sensitivity to temperature changes and are often used as temperature sensors. Thermistors can be of two types: Negative Temperature Coefficient and Positive Temperature Coefficient. For their high accuracy and cost effectiveness, they are widely used in body temperature measurement, temperature monitoring of engines and other important parts, temperature regulators in parts of ovens, refrigerators, and other appliances, industry temperature control, time delay, and limitation of current surge in electrical circuits.

Resistor

A resistor is a basic passive element. It acts as an electrical resistance that is fixed and is used to limit or control the current in electronic circuits as per Ohm's law ($V = I \cdot R$). It is constructed of materials such as carbon, metal films, or metal alloys. There are two types of resistors—fixed and variable. The resistance value is usually displayed in Ohms (Ω).

2.4 Software of the IV System and Infusion Pump

PlatformIO

PlatformIO is a professional-grade, open-source, cross-platform, and cross-architecture Integrated Development Environment (IDE) for IoT. It supports various microcontroller units (MCUs), like ARM Cortex, MSP430, various development boards, frameworks, and environments (Arduino, ESP, and many others). PlatformIO eliminates the low-level hassle of embedded toolchains and integrates the coding to deployment workflow. Project configuration file `platformio.ini` is a declarative document for specifying project work, with complete board management, build flags, and library dependency control.

Web Interface Development

Flutter Web was used to build the web interface, which was then run in the Chrome browser via the Flutter Chrome daemon. Flutter is chosen for its ability to create an efficient and visually effective interface for real-time data visualization, as well as its responsiveness and modern design.

2.5 Working Principle of Automatic Infusion System

This system works hand in hand with an infusion pump that is electronically controlled to regulate the speed of fluid administration and automatic shut-off when there is backflow. The infusion pump acts as the actuation subsystem in a closed-loop control system. The block diagram of the system is shown in Fig. 2.6.

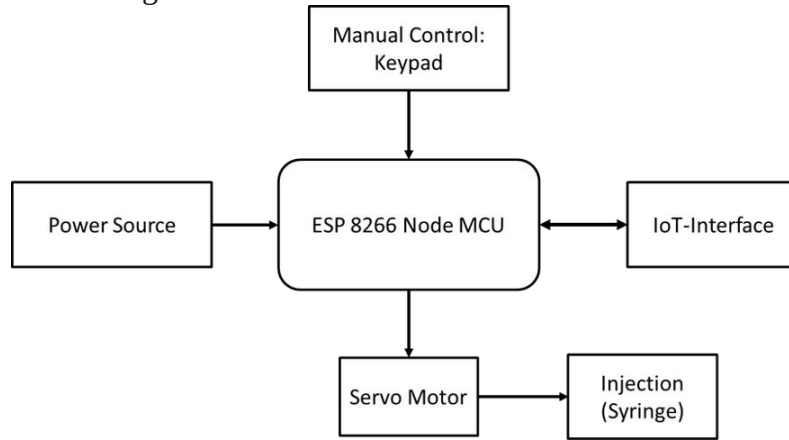


Fig. 2.6. Block diagram of the proposed automated infusion system.

Motion Conversion Mechanism:

The proposed device utilizes a servomotor with a rotation range of 0 to 180 degrees to convert rotational motion into a linear push, allowing the syringe to inject up to 5 mL of fluid, drug, or medication per cycle. Therefore, it's necessary to calculate the degree of rotation of the servo motor from milliliters (mL) unit conversion to make the device more user-friendly.

$$X = \frac{D}{M} \times N \dots\dots\dots(2)$$

Here,

X = Degree of rotation for targeted fluid push through the syringe,

D = Maximum degree of rotation of the Servomotor,

N = Targeted drug push through the syringe,

M = Maximum drug push through the syringe (5mL)

Following equation (2),

For N = 1 mL, $X = \frac{180}{5} * 1 = 36$ -degree rotation of the servo motor.

For N = 2 mL, $X = \frac{180}{5} * 2 = 72$ -degree rotation of the servo motor.

For N = 3 mL, $X = \frac{180}{5} * 3 = 108$ -degree rotation of the servo motor.
For N = 4 mL, $X = \frac{180}{5} * 4 = 144$ -degree rotation of the servo motor.
For N = 5 mL, $X = \frac{180}{5} * 5 = 180$ -degree rotation of the servo motor.

2.6 Hardware of the Smart Infusion Pump

Arduino Mega

The high-performance microcontroller board Arduino Mega is the main control unit for the smart infusion pump system. It accepts inputs from the keypad and sensors and gives control to output devices like the servo motor and solenoid valve. The board contains several digital and analog input/output pins, which allow a wide range of electronic modules to be easily connected. It has a proven processing ability and flexible architecture, ideal for biomedical and automation applications.

Servo Motor

Rack and pinion is coupled to the servo motor, which is employed to accurately move the syringe. Delivers precise angular movement for controlled and consistent fluid delivery during infusion.

4×4 Keypad Module

The user input and parameter setting are done using the 4×4 keypad module. It helps the user to efficiently manage system functions and adjust infusion parameters.

MB102 Breadboard Power Supply Module

The MB102 contains a power supply module that supplies 5 V regulated power to the Arduino Mega and peripheral devices. It guarantees a steady voltage level in the system, which is crucial for proper functioning.

Chapter 3

Implementation

3.1 Smart IV system

The power of the IV system was delivered by a 3.7V Li-Po battery, which then needed to be stepped down to 3V by an MP1584EN step-down module to support the ESP32-S2 Mini module. The thermistor and the MAX30102 Pulse and Heart Rate Sensor were connected to the ESP32-S2 Mini as a sensing unit. An LED and LM393 LDR sensor module were connected to the ESP32-S2 Mini. An IRFZ44N N-channel MOSFET was used for switching the solenoid valve, and a 1N4007 Diode was used as a flyback diode, which protects other circuit components from voltage spikes that occur when the solenoid valve is switched off. A 3V output connection was taken from the ESP32-S2 Mini microcontroller to the MT-3608 DC-DC Step-up Module to boost the voltage to 3.7 and power the solenoid valve. A TP4056 Lithium Battery Charger Module was connected to the battery for charging the battery using a Micro USB-B cable. The entire circuit was packed inside a 3D-printed dual-chamber miniature case. That small case works as the complete device. The circuit diagram is shown in Fig. 3.1.

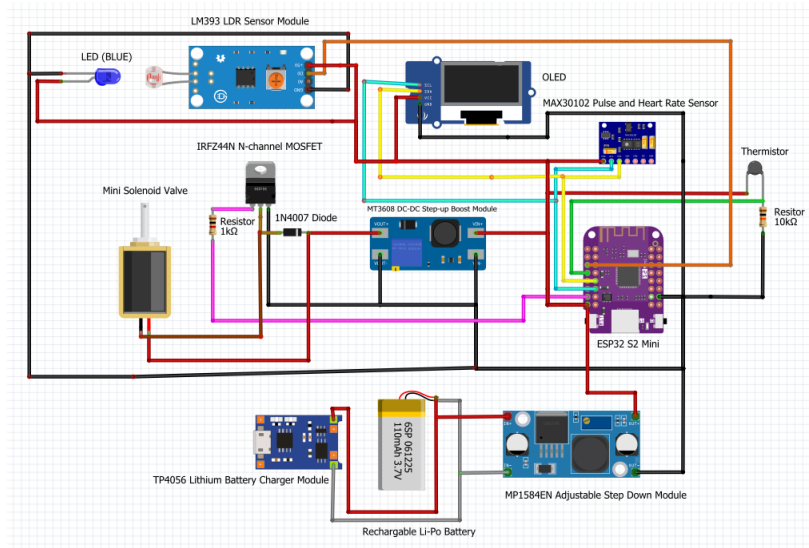
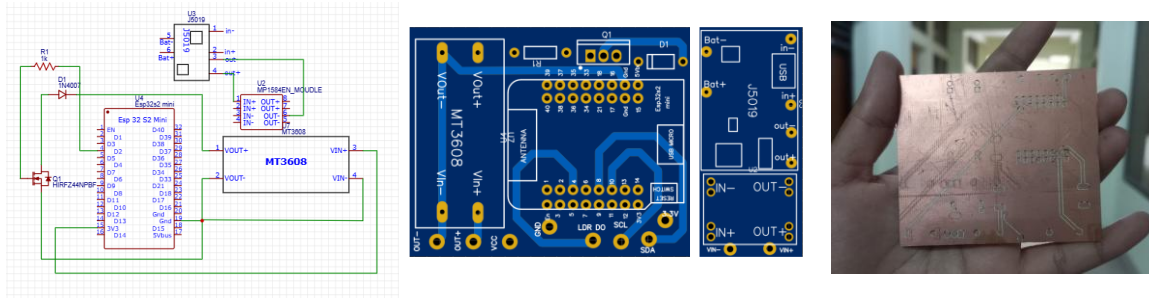


Fig. 3.1: Circuit diagram of the IV system

After designing a circuit layout and schematic using EasyEDA, the PCB fabrication process was carried out. A double-layer PCB design was chosen for compactness, structured routing, and to ease the complexity of wiring. To guarantee electrical connectivity and reduce noise interference, a minimum track width of 1.5 mm and clearance spacing of 0.5 mm were kept during layout optimization. The PCB fabrication process is shown in Fig. 3.2. The finalized PCB layout was then manufactured on a copper-clad board, which was then drilled, and the components were placed and soldered to get the complete hardware prototype of the proposed smart IV cannula system.



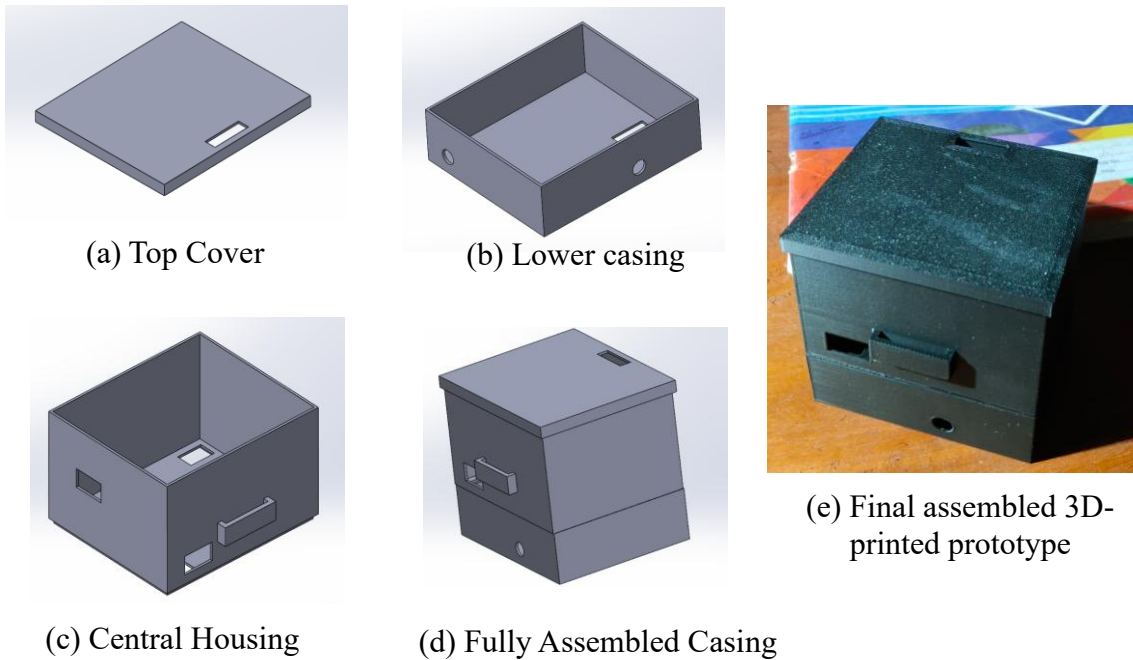
(a) Schematic Design

(b) 2D view of the developed PCB layout

(c) Printed Routing on Copper Clad Board

Fig. 3.2: PCB fabrication (a) Schematic Design, (b) 2D view of the developed PCB layout, (c) Printed Routing on Copper Clad Board

The CAD model of the enclosure for the proposed system was created in SolidWorks, considering the finalized dimensions of the PCB and the measurement of the IV cannula. The prototype was 3D printed with black PLA 3D printing to provide a lightweight, durable, and portable chassis for the created system. The CAD parts are shown in Fig. 3.3. The lower casing was designed to fit the LDR sensor, LED, and the thermistor module, and the saline inlet–outlet tube for the solenoid valve. The top cover was designed to mount the OLED display for real-time monitoring, and the central housing section housed the PCB, battery, and associated control circuitry.



(a) Top Cover

(b) Lower casing

(c) Central Housing

(d) Fully Assembled Casing

(e) Final assembled 3D-printed prototype

Fig. 3.3: 3D model for casing (a) Top Cover, (b) Lower casing, (c) Central Housing, (d) Fully Assembled Casing, (e) Final assembled 3D-printed prototype

3.2 Web Interface of Smart IV Monitoring System

The web interface was built using Flutter Web through Android Studio. Using the web server, our system can be controlled both automatically and manually. The dashboard showed an interruption in normal infusion flow, as shown in Fig. 3.4 when an abnormal flow condition is detected.

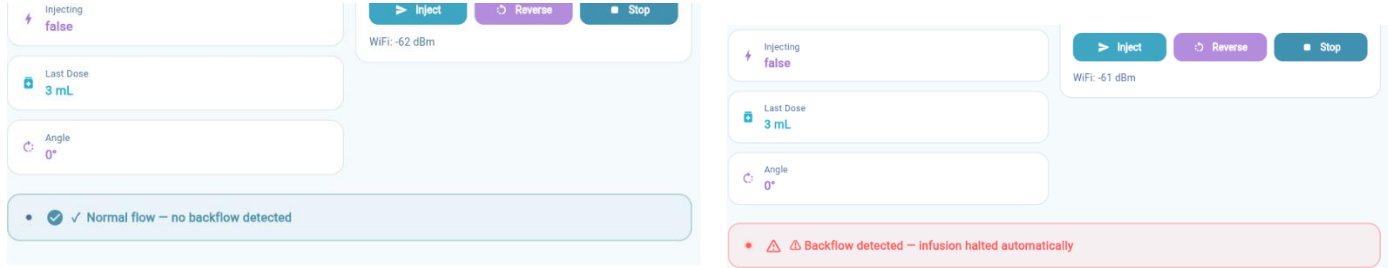


Fig. 3.4: Smart Infusion Monitoring Dashboard interface under different health conditions

3.3 Smart Infusion Pump

For controlled fluid delivery in the smart infusion pump module, the rotational motion of the servo motor was translated into linear motion with the help of a rack and pinion mechanism as illustrated in Fig. 3.5. The dimensional specification of the syringe and actuator assembly is used to design the mechanical components on SolidWorks, such as rack and pinion body, syringe holder, support bases, and alignment structures. PLA material was used for 3D printing the CAD parts. The pinion gear was directly connected to the servo motor shaft, allowing the rack to be displaced in a linear manner to apply a controlled force and controlled infusion rate to the syringe plunger. Other supporting bases and guiding elements (as shown in Fig. 3.4) were added to ensure the stability of the syringe, the alignment, and smooth linear movement of the syringe during operation.

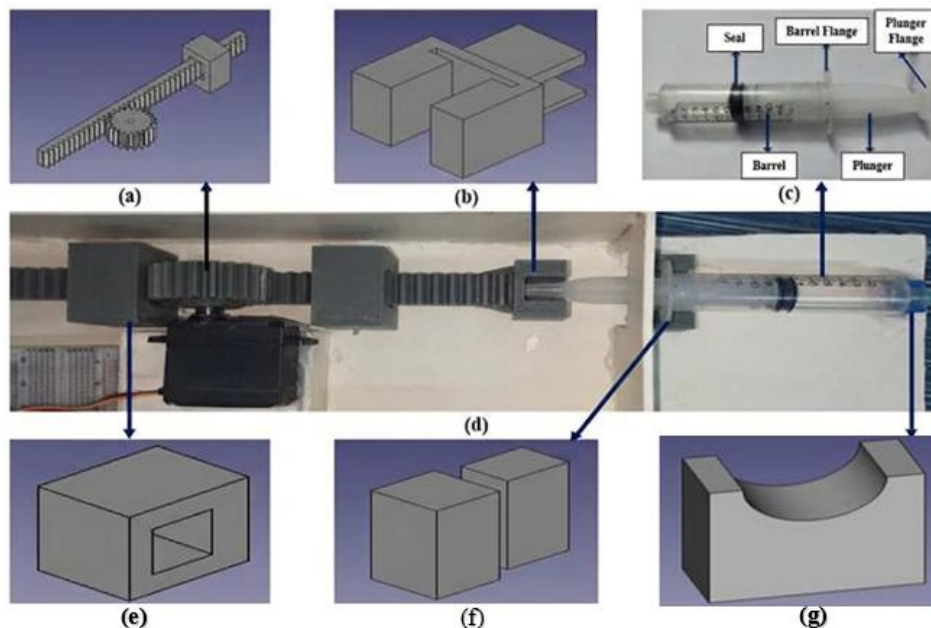


Fig. 3.5: (a) Rack and pinion body, (b) base-1, (c) different parts of a syringe, (d) complete assembly of the 3D model to create the syringe pump, (e) base 2, (f) base 3, and (g) base 4

An Arduino Mega was used for the smart infusion pump system's control circuit. The microcontroller was connected to a servo motor for accurate syringe actuation and fluid delivery via the rack-and-pinion mechanism. A 4×4 keypad module was also built into the unit for input from the user and infusion parameter control. To maintain stable system operation, the supply of regulated 5 V was provided by an MB102 breadboard power supply module to the Arduino Mega, servo motor, and peripheral circuitry. The complete circuit configuration is as shown in Fig. 3.6.

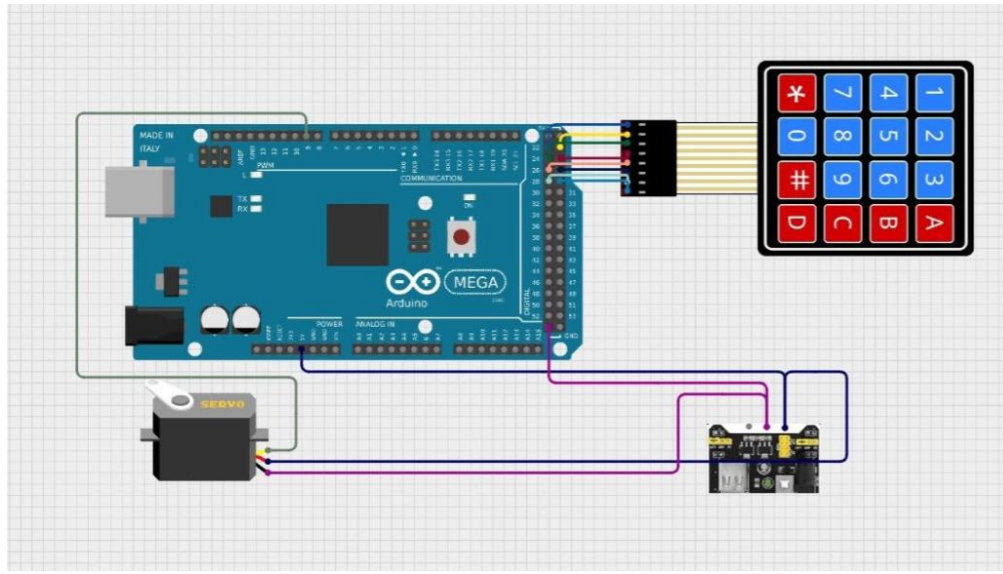


Fig. 3.6: Circuit diagram of infusion pump control system

The final assembled system of the smart infusion pump, including the syringe actuation mechanism, servo-driven rack and pinion, control circuit, and user input interface, is housed in a compact enclosure as shown in Fig. 3.7. The final prototype was tested to confirm the stability of the syringe movement and the operation of the fluid delivery.



Fig. 3.7: Assembled Infusion pump

3.4 Web Interface Integration of Infusion Pump

In this project, Flutter was used to build a browser-compatible user interface through Android Studio. The interface helps to communicate with two ESP32-based modules over a local WiFi network. The dashboard periodically fetches data from the ESP32 servers every two seconds using HTTP GET requests. The received information was encoded in JSON format, decoded within the Flutter interface, and displayed dynamically on the screen. The interface in Fig. 3.8 shows live updates of heart rate, SpO₂, temperature, valve condition, flow status, automatic mode status, dose activity, injector status, and other system parameters. For an abnormal flow condition such as a backflow or flow stoppage, the dashboard automatically displays a warning message.

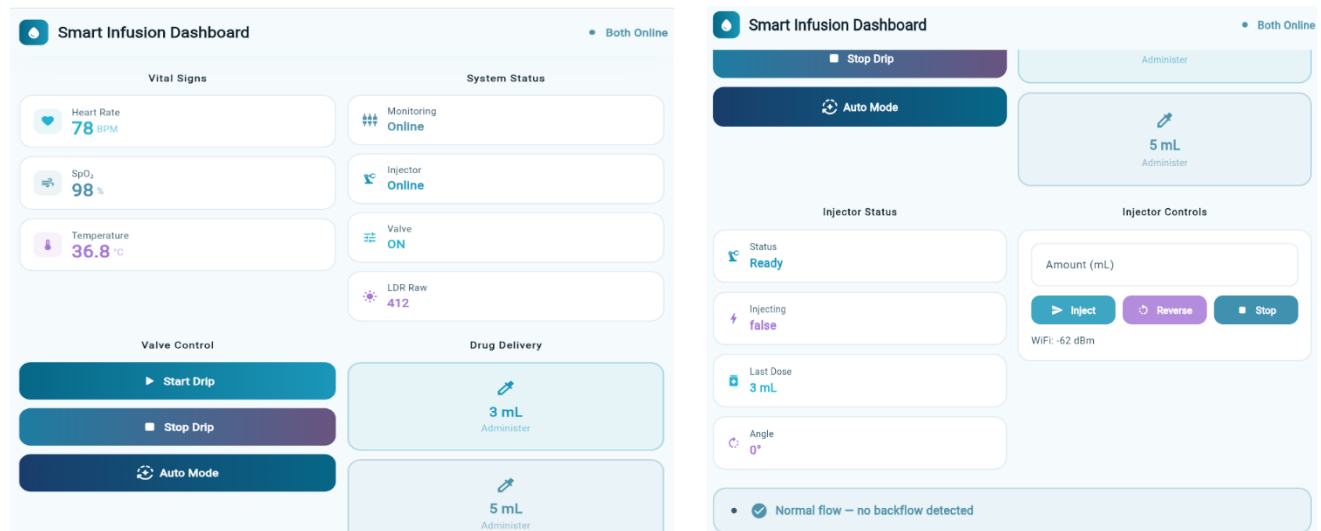


Fig. 3.8: Smart Infusion Monitoring Dashboard interface under normal health conditions

Chapter 4

Result and Discussion

4.1 Smart IV System

The development of the Reverse Blood Flow Prevention System has already been completed, as shown in Fig. 4 .1. The detection circuit and control valve circuit of reverse flow have been completed, such that any case of reverse flow in the IV line would be automatically detected, and the valve will close immediately to prevent potential contamination or clotting. Sensor calibrations and validations have already been done for all sensors to verify the performance of the circuitry and its response in detecting reverse flow in different flow rates of saline solution. This step ensures the basic functionality of the proposed system before moving forward to integrating other modules.

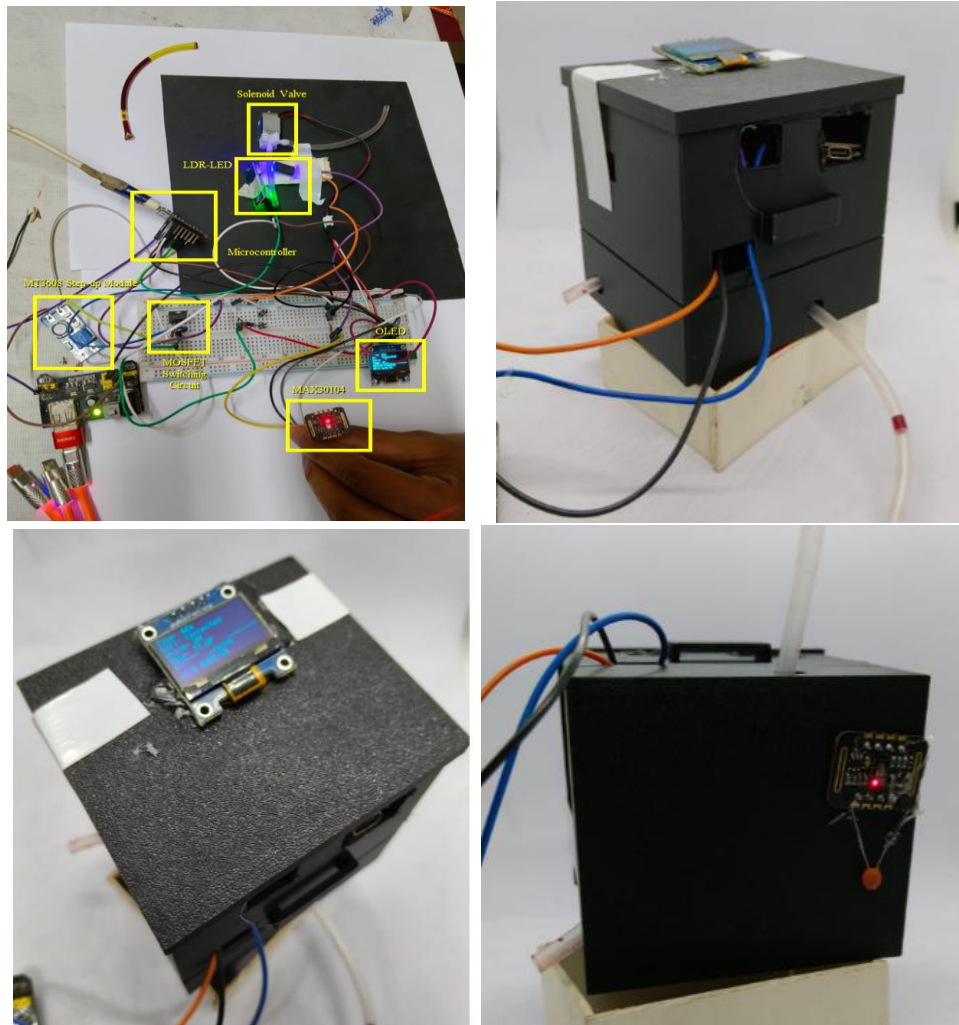


Fig 4.1: Assembled IV System

The device performed precisely and stopped the backflow of blood whenever it was about to occur. The LDR sensor has achieved satisfactory results. Blood typically exhibits higher absorption of blue LED light. Therefore, the presence of blood interrupted the light path between the LED and the LDR sensor, leading to a higher sensor value that goes beyond the predefined threshold. Along with that, the solenoid valve got closed and prevented the backflow with the help of an automated locking system.

The OLED displayed the data according to the provided instructions. All the measured physical parameters (body temperature, SpO2, and Heart Rate) were presented in the display of the OLED, including some more information such as the elapsed time since activation, the status of the Wi-Fi connection, and the current state of the valve. After the valve gets turned off, the OLED displays a “Call the nurse” message.

4.2 Results of the Smart IV System

Table 4.1: Performance Checking of LED -LDR Setup
(1 = Blood Detected; 0 = Not detected)

No.	Manual Observation	Detection
1	0	0
2	0	0
3	0	0
4	0	0
5	1	1
6	1	1
7	1	1
8	1	0
9	1	1
10	0	0
11	0	0
12	1	1
13	1	0
14	1	1
15	1	1
16	0	1
17	1	1
18	1	1
19	1	1
20	1	1

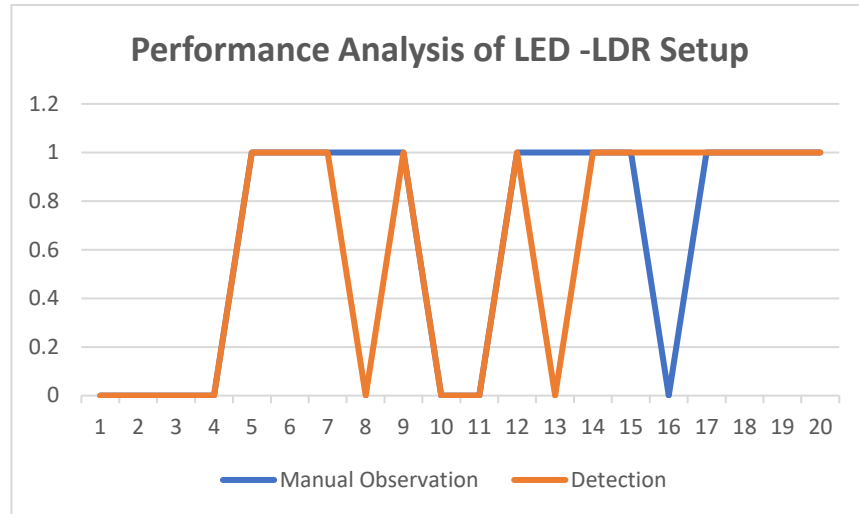


Fig. 4.2: Performance Curve of LED-LDR Blood detection (X axis = Observation; Y axis = value)

$$\text{Accuracy} = \frac{\Sigma \text{ Matched Value}}{\text{Total number of values}} * 100\% = \frac{17}{20} * 100\% = 85\%$$

Table 4.2: Response time for solenoid valve to close after blood detection (charted by keeping on/off time using a stopwatch manually)

No	Time (s)
1	0.44
2	0.43
3	0.46
4	0.38
5	1.2
6	0.56
7	0.40
8	0.86
9	0.42
10	0.57

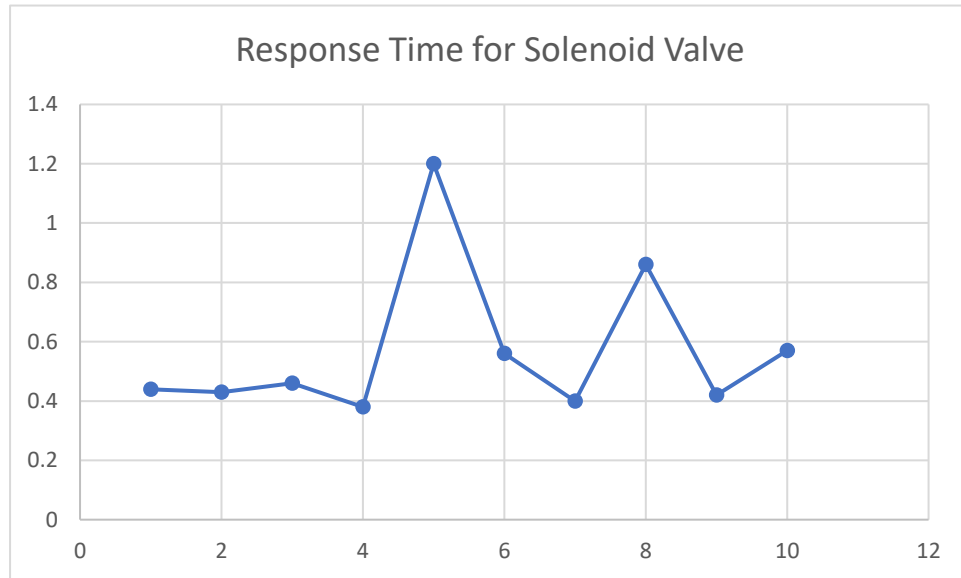


Fig. 4.3: Response time of the solenoid Valve to close after blood detection
(X axis = Observation; Y axis = value)

4.3 Web Interface of Smart IV Monitoring System

The dashboard showed an interruption in normal infusion flow, as shown in Fig. 4.4 (b) when an abnormal flow condition is detected.

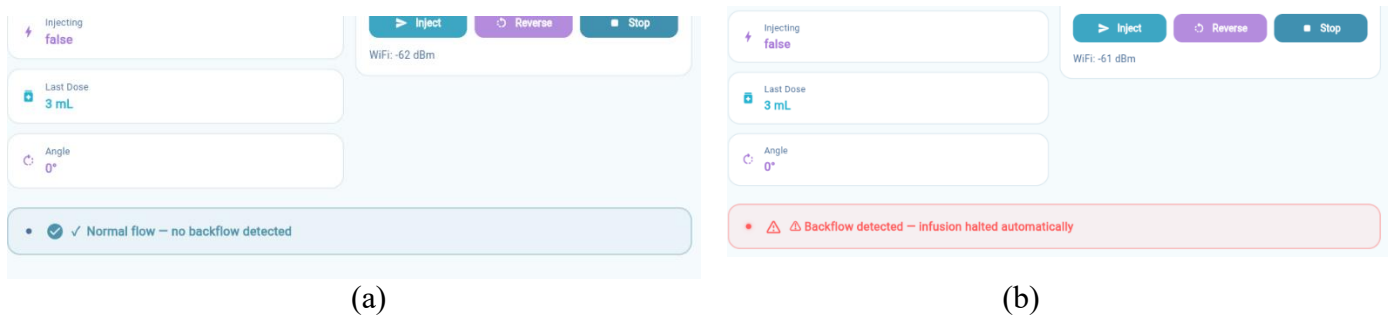


Fig. 4.4: Smart IV Monitoring Dashboard under (a) normal and (b) abnormal flow conditions

4.4 Smart Infusion Pump

The main objective of this study was to develop a smart infusion pump capable of reducing errors found in existing systems. The syringe pump utilized a servo motor with a rack-and-pinion mechanism to convert rotary motion into linear motion, allowing controlled medication injection. Additionally, the web interface enabled users to set the dosage amount and monitor both the injected and remaining medication volume in milliliters (mL), as shown in Fig.4.5.



(a)



(b)

Fig. 4.5: (a) Initial syringe position and (b) syringe position after injection controlled through the web interface

4.5 Result of the Infusion Pump

Table 4.3. Flow rate determination with the required time

Trial Number	Medication Injected (mL)	Required Time in seconds	Flow Rate (mL/sec)
1	3	3.70	0.81
2	5	5.70	0.88
3	7	8.24	0.85
4	10	12.80	0.78
5	13	15.67	0.83
6	15	17.60	0.85

Average Flow Rate = 0.83 mL/sec

The comparison analysis in Table II demonstrates that the proposed IoT-based syringe pump achieved a flow rate of **0.83 mL/sec** with an accuracy of approximately **99%**. Compared with previously reported syringe pump systems, the developed device showed improved precision, efficient wireless control, and reliable medication delivery performance. The integration of IoT technology, servo motor control, and rack-and-pinion mechanisms contributed to reducing medication delivery errors and enhancing automated drug administration efficiency.

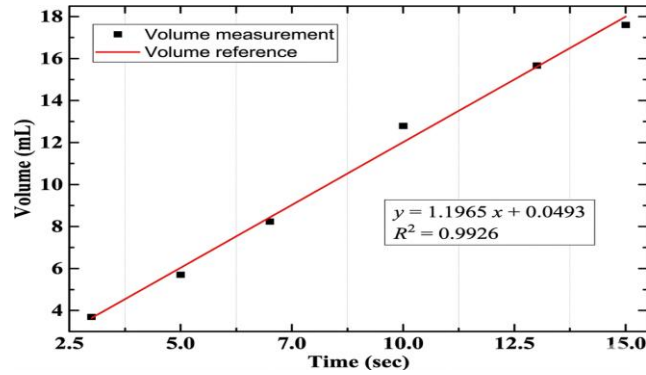


Fig. 4.6: Linearity evaluation of the IoT-based syringe pump, showing the relationship between infusion time and plunger volume

Fig. 4.6 illustrates the linearity evaluation of the IoT-based syringe pump by plotting infused medication volume against injection time. The graph shows a strong linear relationship between infusion time and plunger displacement, with the regression equation:

$$y = 1.1965x + 0.0493$$

The high linear correlation confirms that the syringe pump provides stable and consistent fluid delivery with minimal deviation during operation. This indicates that the servo motor and rack-and-pinion mechanism successfully achieved precise linear motion control for accurate medication infusion.

4.6 Automated Drug Delivery Web Interface

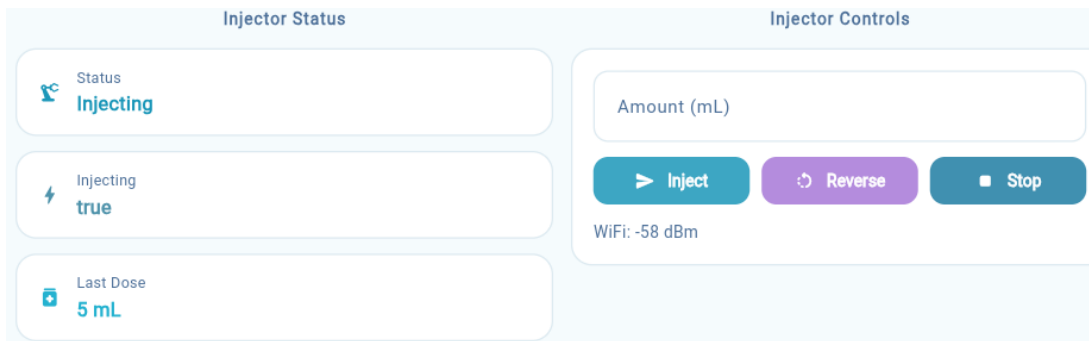


Fig. 4.7: Injector status while injecting a 5 mL dose of medicine to the patient as displayed on the web interface

The system has been able to deliver the drug precisely in the correct amount without manual control, as shown in Fig. 4.7. Along with that, the system can be monitored in a hospital environment with its status showing on the dashboard without much manual input.

Chapter 5

Conclusion, Future Plan, and Gantt Chart

5.1 Conclusion

This work presents the design and development of a smart tele-controlled IV system based on an IoT system that incorporates drug delivery functions from an infusion pump with reverse blood flow monitoring for enhanced patient care and for remote health care applications. The system developed here is based on an ESP32 S2 mini microcontroller, an optical sensing technique for detecting reverse blood flow, and a rack-and-pinion mechanism with a servomotor to control the delivery of the fluid with accuracy and regulation. The blood has a greater absorption value than other liquids, and because of that, during back-flow conditions, the value was consistently observed to be higher, resulting in better patient safety and monitoring. The infusion pump was programmed to operate in either milliliter or degree mode with a flow rate of 0.83 mL/s and a mean absolute error (MAE) of $1.42 \pm 0.47\%$, which was acceptable for delivering the infusion. Further, the compact, lightweight, and portable design enhances the ease of use and patient comfort, and the mobile application allows for real-time medication delivery monitoring, control, and scheduling. The proposed system holds great promise as an affordable, effective, and intelligent solution to the present-day automated intravenous therapy and telehealth medical system.

5.2 Limitations and Future Plan

The proposed system was found to be reliable, but there were some restrictions for its clinical application. The reverse blood flow monitoring section is currently based on inexpensive sensors with no air embolism detection, and the infusion pump section has a maximum of 5 mL per injection. Medical-grade sensors, air bubble detection, IV drip and volume monitoring, and improved monitoring via the IoT for greater accuracy, safety, and telehealth usability are all possibilities for future enhancements.

Table 5.1: Gantt Chart of Our Work Plan (Weekly Basis)

Task \ Week	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13
Proposal approval									
Component collection									
Circuit design and assembly									
Progress presentation									
Circuit assembly (final)									
CAD design									
Final presentation slide making and report writing									
Project showcasing									

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